



University of Electronic Science and Technology of China

电子科技大学

TDPS Final Presentation

Team 15 PTSD

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CONTENTS



Project Outcome



General Design



Challenges



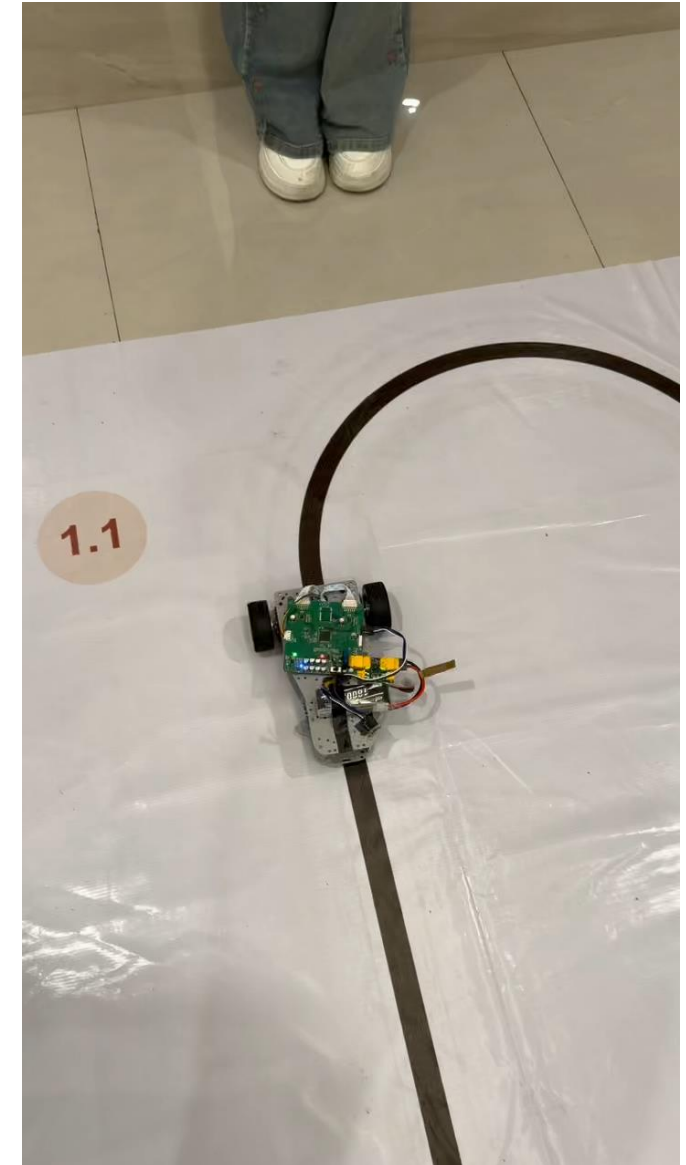
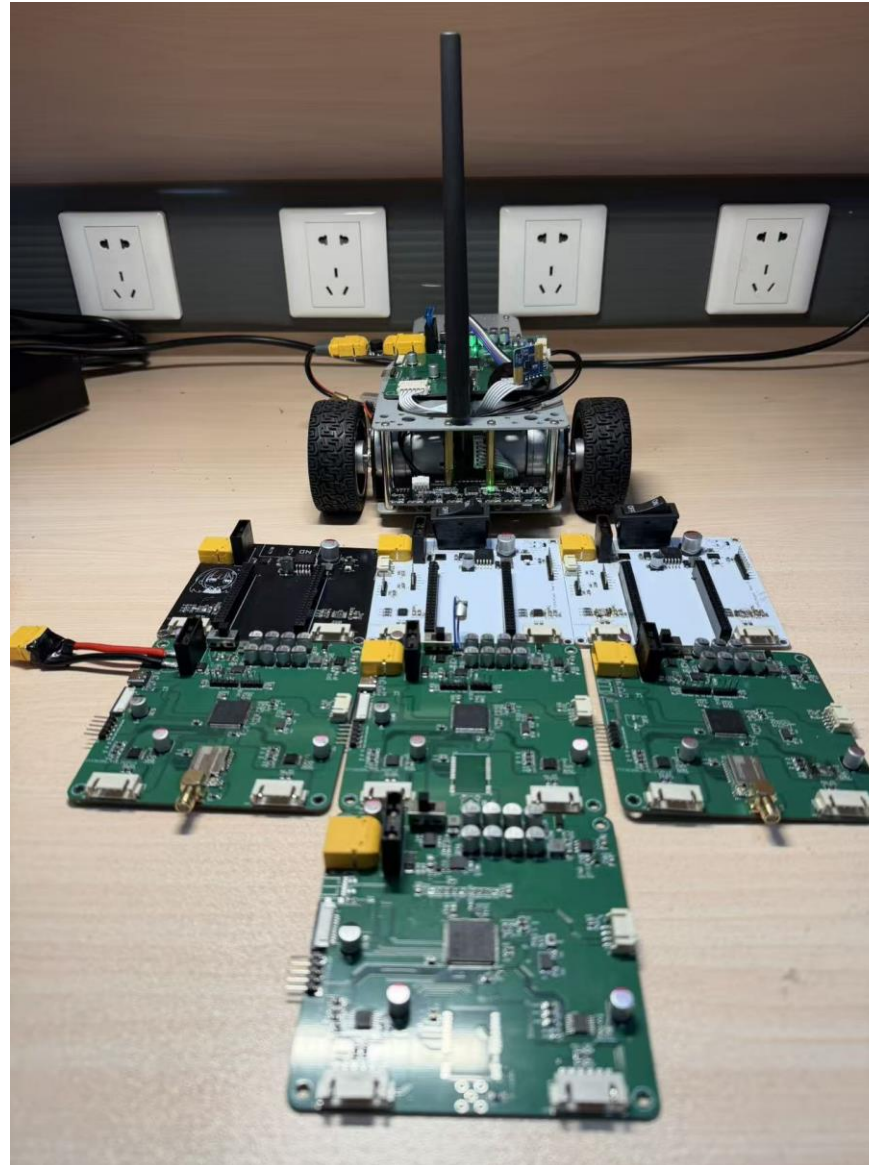
Module Details

Final Demo Results:

- ✓ Line following
- ✓ Obstacle detection
- ✓ LoRa communication

Other Outcomes:

-  Special designed PCB
-  Reusable software solution

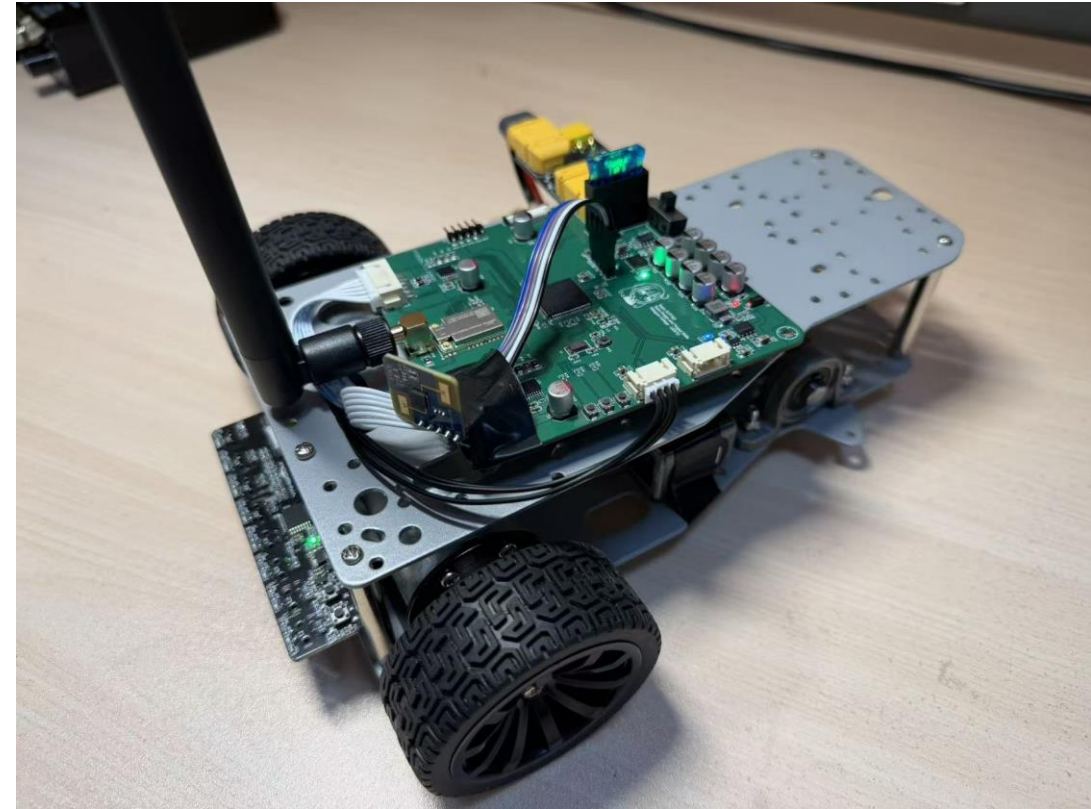


Chassis Selection:

- Differential tricycle: easy to design and test
- Inverted tricycle: more flexible but not stable
- Forward tricycle: very hard to finish line following

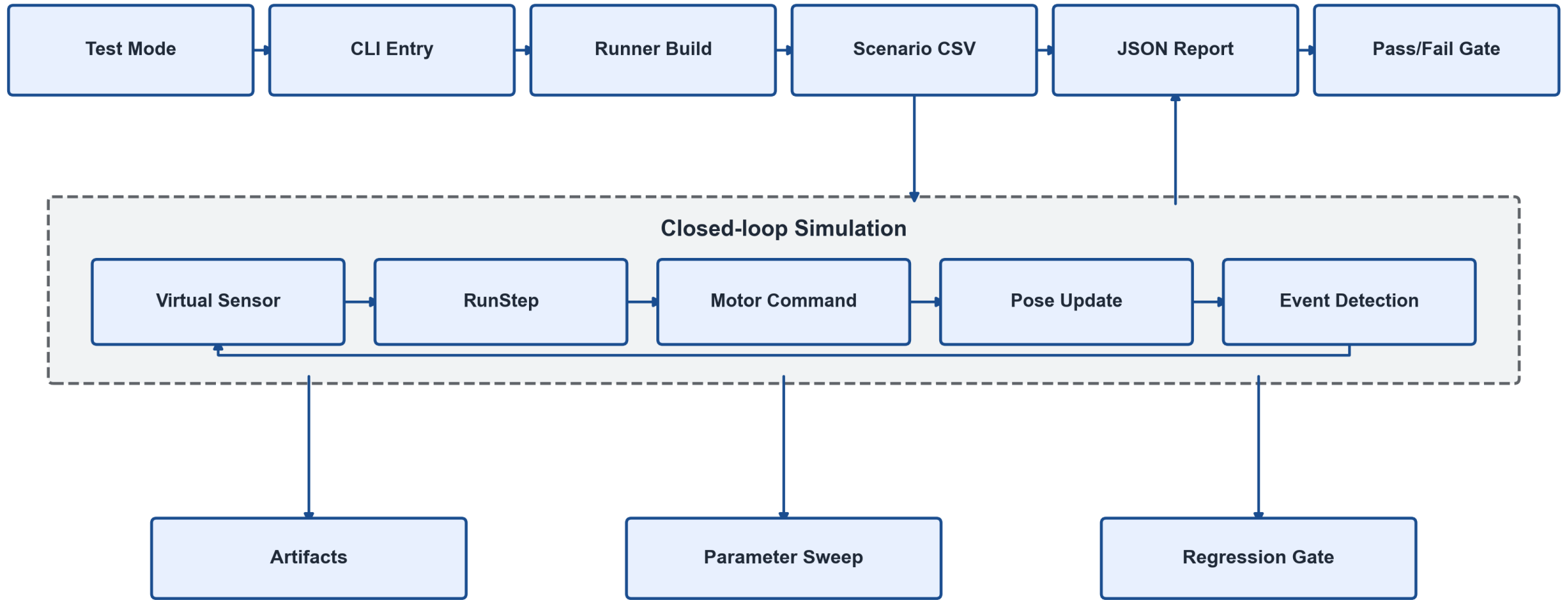
PCB Design:

- Adapter board: easy to design, low reliability, problems with power supply
- Integrated PCB: much smaller, more stable, much better power supply



Module	Type
Line-following Sensor	Yahboom 8-LP Sensor
LoRa Module	EWT22A-900BWL22S
Radar	HLK-LD2410S
MCU	STM32F407VET6

Simulator Regression Architecture

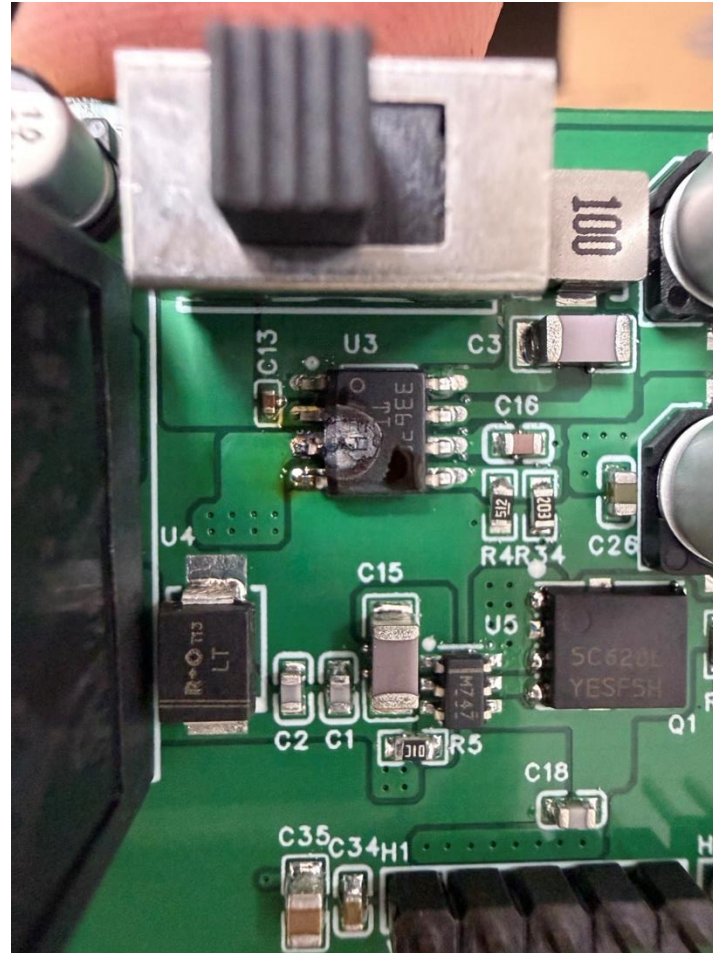


Challenges:

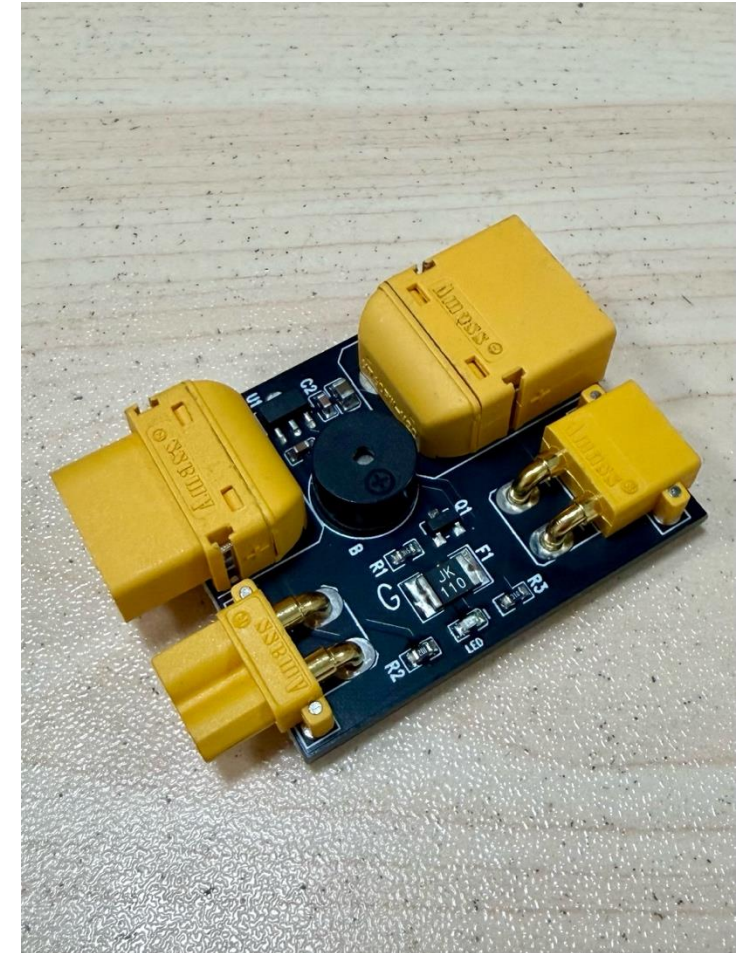
- Too difficult PCB design
- Power supply problems
- Real test is different from simulator

Insights:

- Design and test step by step
- Add battery protection module
- Add noise and disturbance into the simulator



A broken board



Battery protection module

Line Following Strategy & System Design

① Design Objective



Core Requirements



Robust line tracking



Stable cornering performance



Real-time steering adjustment



Robust operation on different track sections



Challenges

High speed



Oscillation and overshoot

Sensor noise



Unstable steering commands

Straight require speed

VS

Curves require stability

② System Architecture

Line Sensor Array



Position Detection



Error Calculation



PID + kff Controller



Differential Steering



Motor Driver



Vehicle Motion

③ Design Highlights



Modular
Architecture



Speed-Adaptive
Control



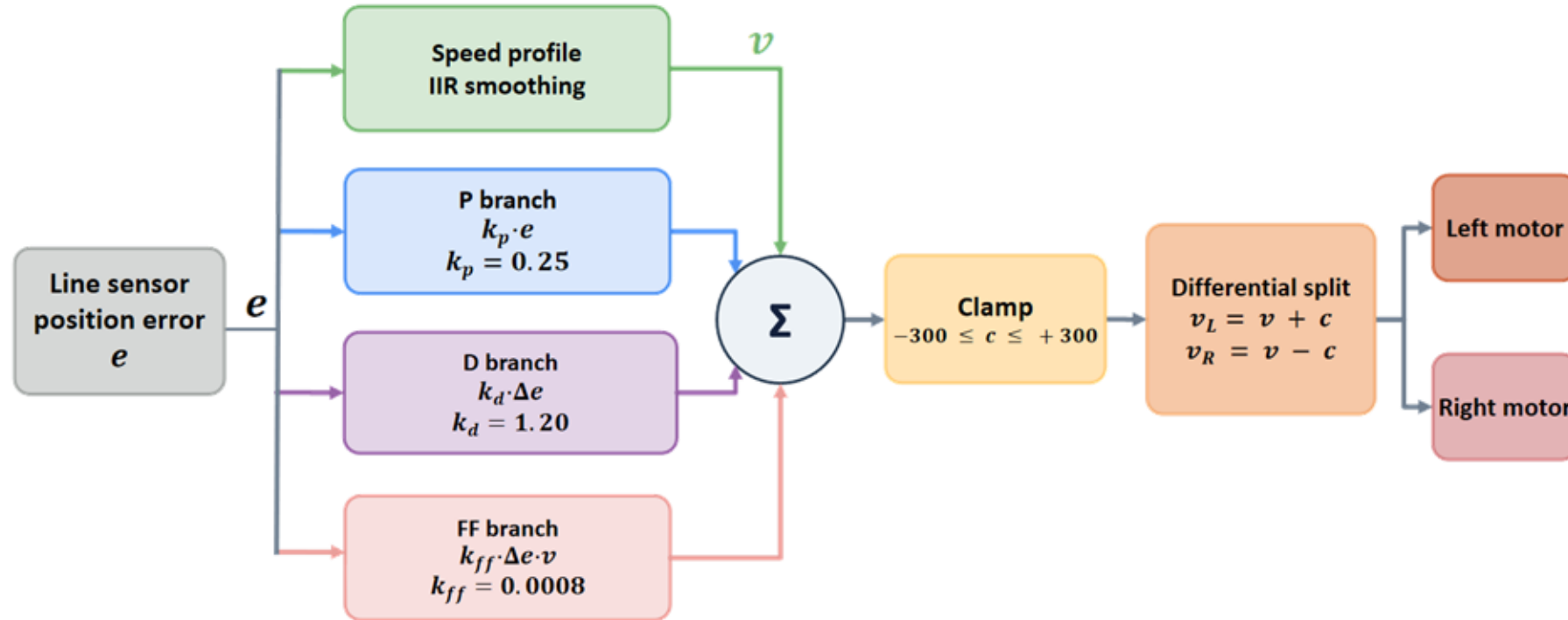
Differential
Steering



Real-Time
Execution

① Controller Structure

$$c = \text{clip}(k_p \cdot e + k_d \cdot \Delta e + k_{ff} \cdot \Delta e \cdot v, \pm 300)$$



② Controller Strategy

P Branch
 $u_P = K_p \times e$

D Branch
 $u_D = k_d \times \Delta e$

FF Branch
 $u_{FF} = K_{ff} \times \Delta e \times v$

$$v_L = v + c$$

$$v_R = v - c$$

③ Why PD+kff ?

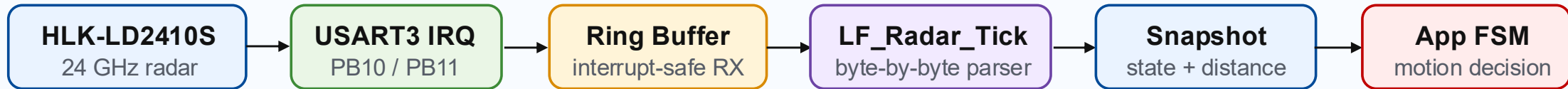
Control Method	Characteristics
P Only	Fast but oscillatory
PD	Stable but slower
PD+kff	Fast and stable

Achieved Benefits

- ✓ Reduced oscillation
- ✓ Faster response
- ✓ Smooth cornering
- ✓ Stable high-speed line tracking

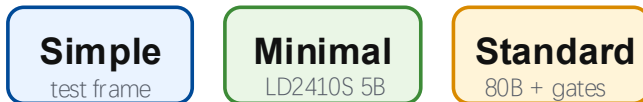
NON-BLOCKING TECHNICAL ROUTE

Radar stays inside the 10 ms loop: interrupt RX, incremental parsing, then motion arbitration from a fresh snapshot.



1. Frame Parser

Supported inputs



- Custom, minimal, and standard frames.
- Distance in mm; gate energy for debug.

2. Decision FSM



Robustness rules

3-frame debounce · 300 ms timeout
450/650 mm hysteresis

3. Avoidance Strategy

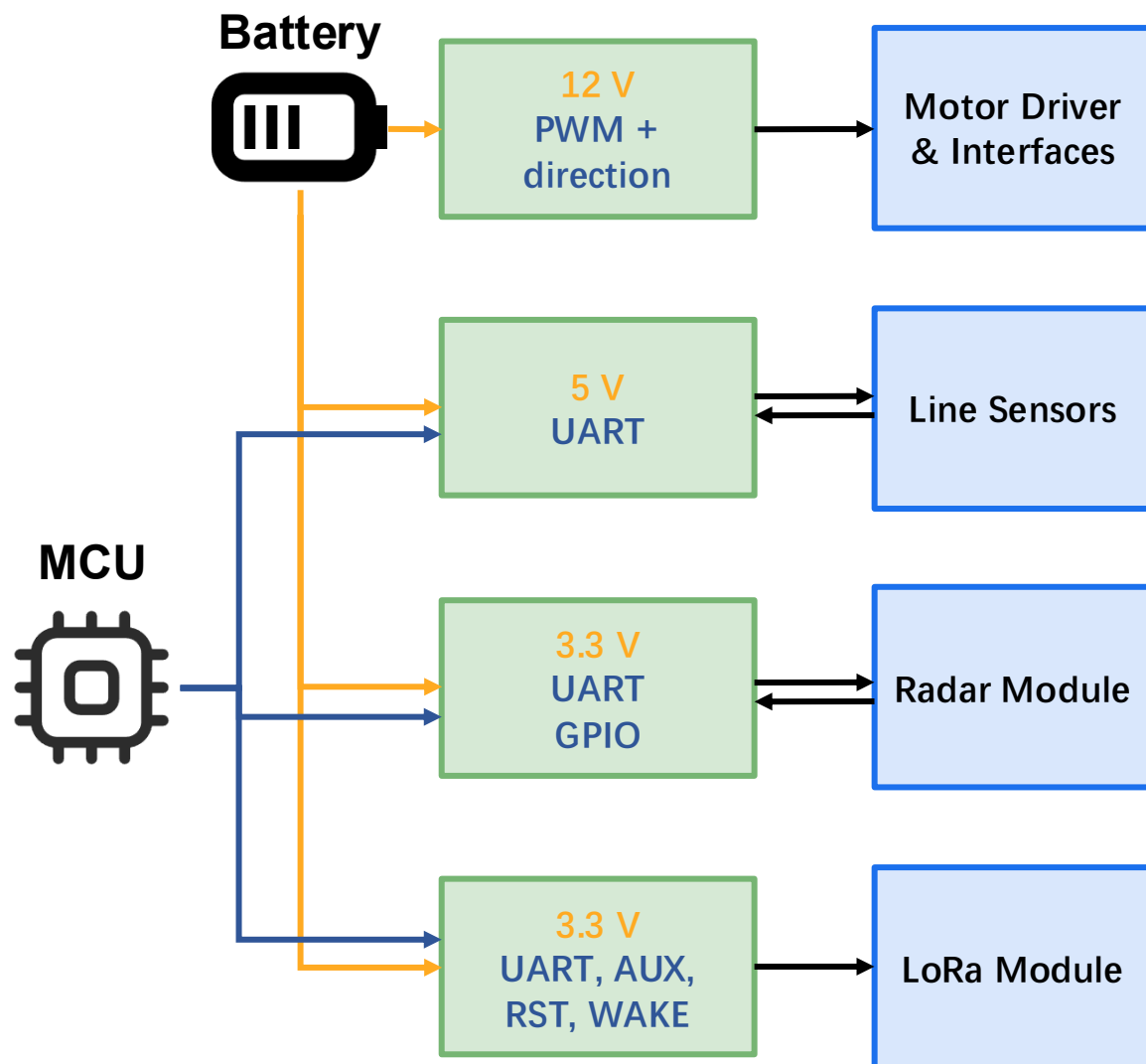
BLOCK response



- WARN slows down; BLOCK triggers bypass

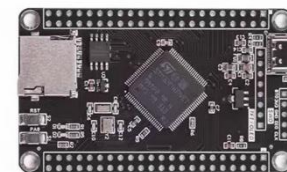
Limit: forward-only radar; heuristic bypass.

PCB Integration: Power and Interfaces

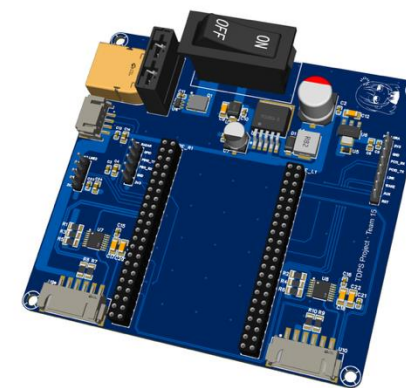


Initial PCB Design:

Adapter board + STM32F407VET6



+

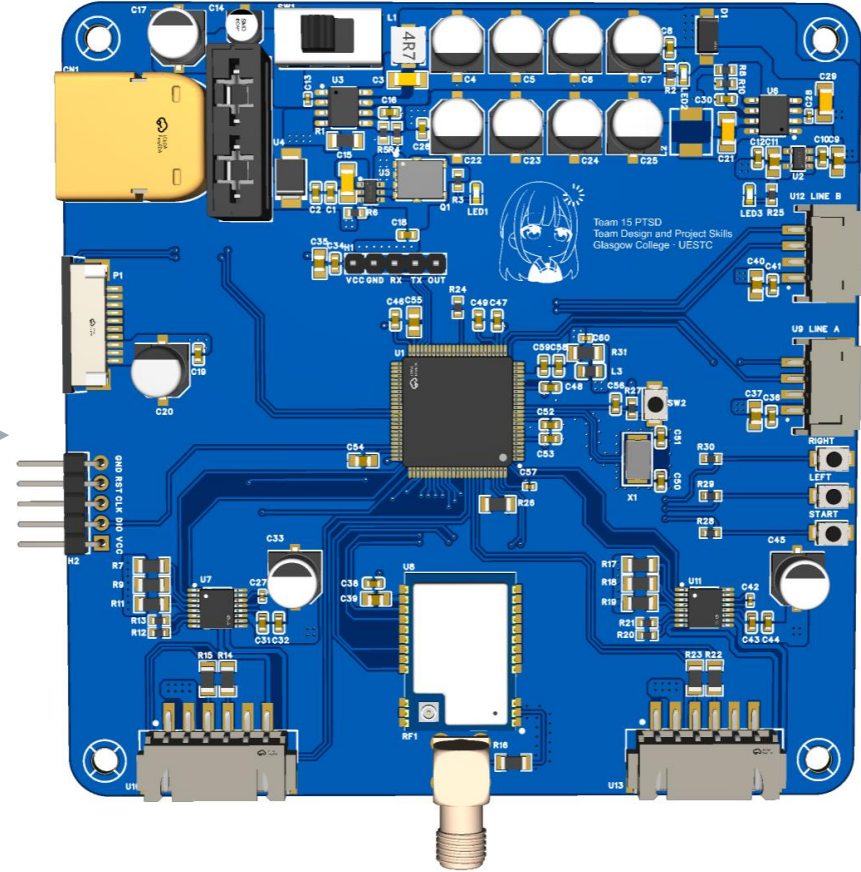
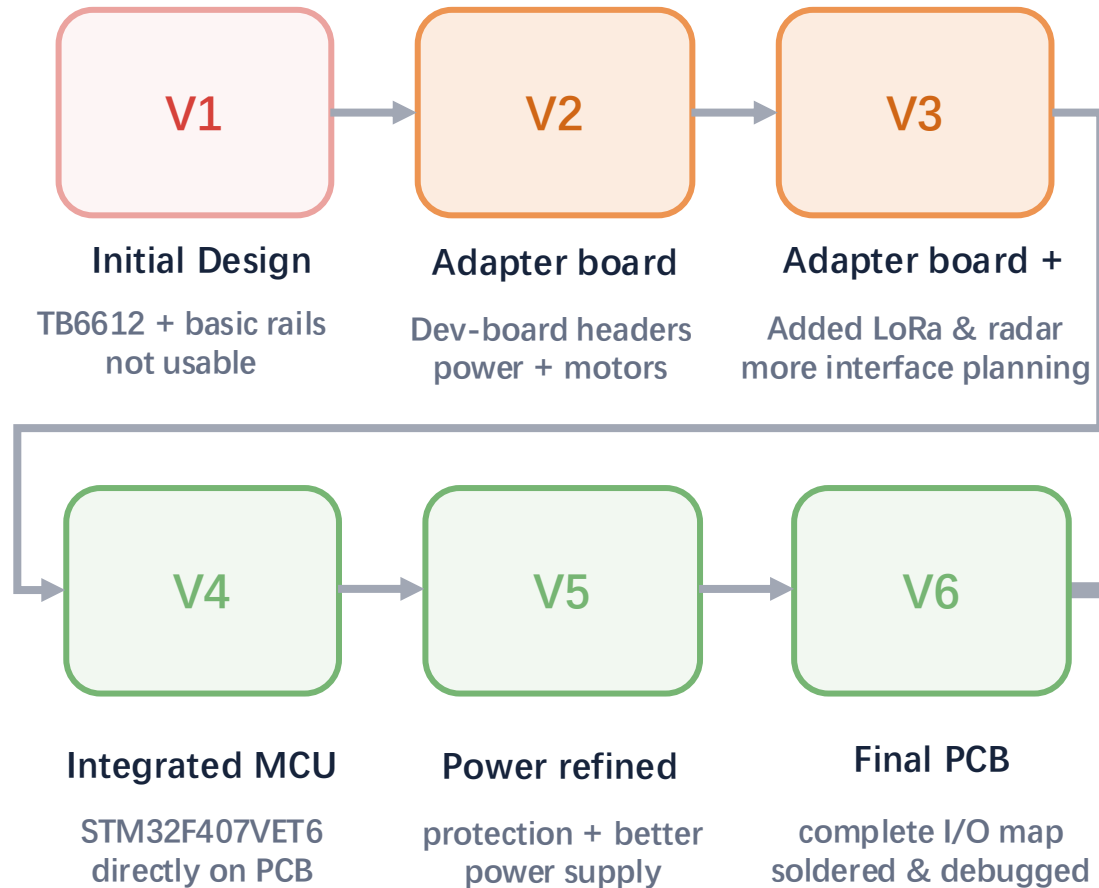


Limitation:

Loose and unreliable wiring

Poor power stability

PCB : Six Iterations to PCB6



PCB6 : Integrate all modules in one

PCB Schematic Overview

Battery input and protection

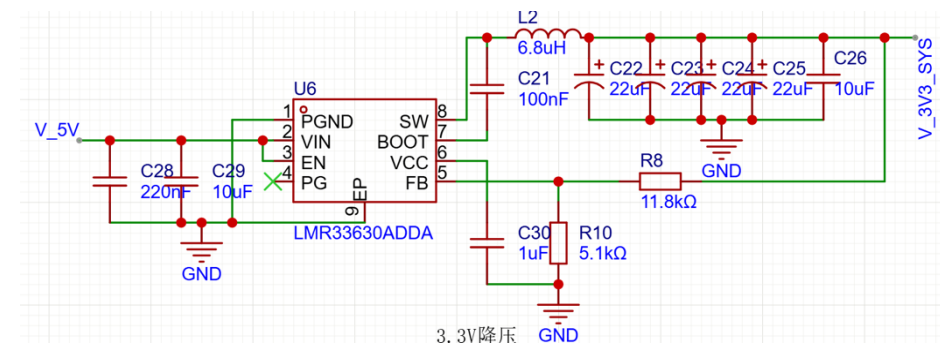
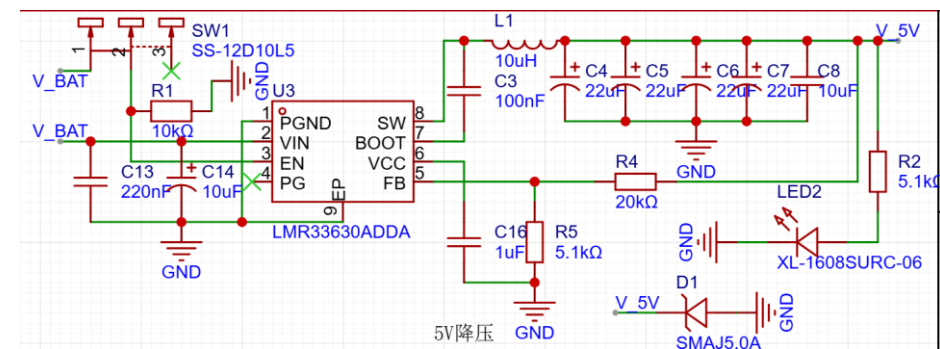
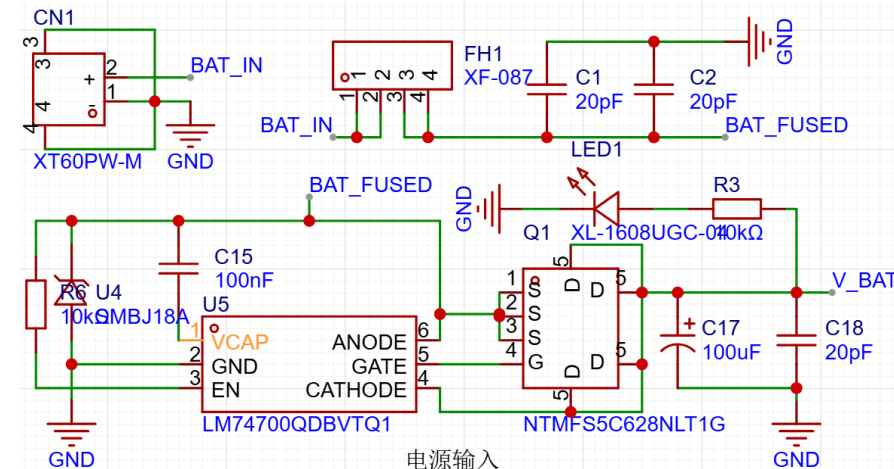
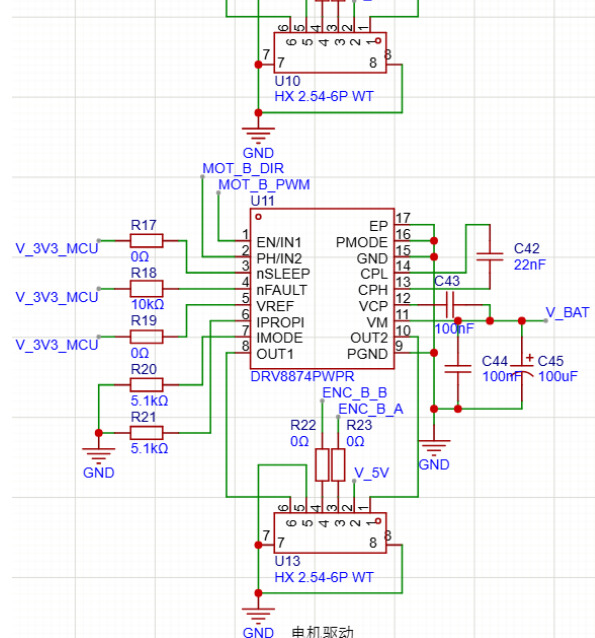
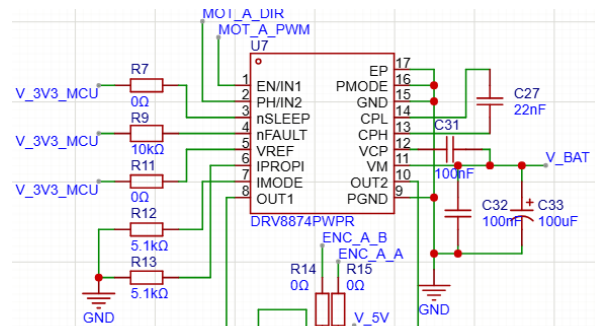
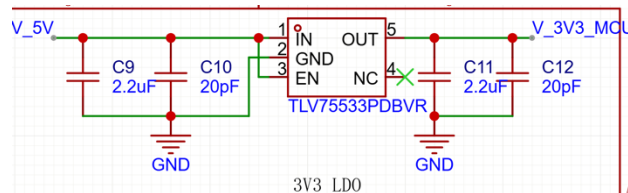
- Battery protection module
- Fuse, ideal-diode and TVS protection
- Reduces reverse-connection and transient risks

Power Supply

- Battery rail → **5 V buck converter**
- 5 V rail → **3.3 V system rail**
- 5 V rail → **3.3 V MCU rail**
- Motor driver: high-current battery path
- MCU and sensors: isolated low-voltage rails

Motor driver integration

- Two DRV8874 motor driver circuits
- PWM and direction signals from MCU
- Bulk capacitors near motor supply pins





Requirement Analysis

The vehicle must automatically report checkpoint information during operation.

The task is decomposed into three key sub-problems:



Checkpoint Detection

Determine when the vehicle reaches a checkpoint



Time Recording

Calculate elapsed time from task start



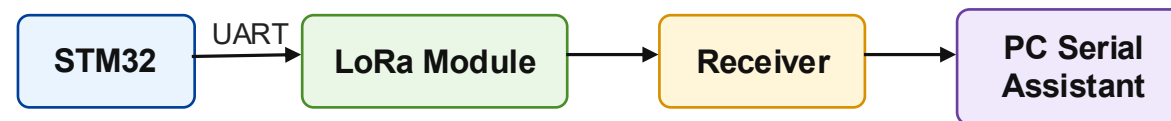
Wireless Communication

Transmit checkpoint information to the receiver



Initial Design (Version 1)

Communication Link



Hardware selection



EWT22A-900BWL22S



Limitation

Solved: How to send data

Missing: When to send data



How can checkpoint events be detected reliably and automatically?

Version 2



HLK-LD2410S-24G

Problems

- ✗ Detects front obstacles only
- ✗ Cannot detect overhead checkpoint gates
- ✗ Unable to provide a reliable trigger event



Key Considerations

Right Sensor
Reliable Trigger
Simple Implementation

Version 3

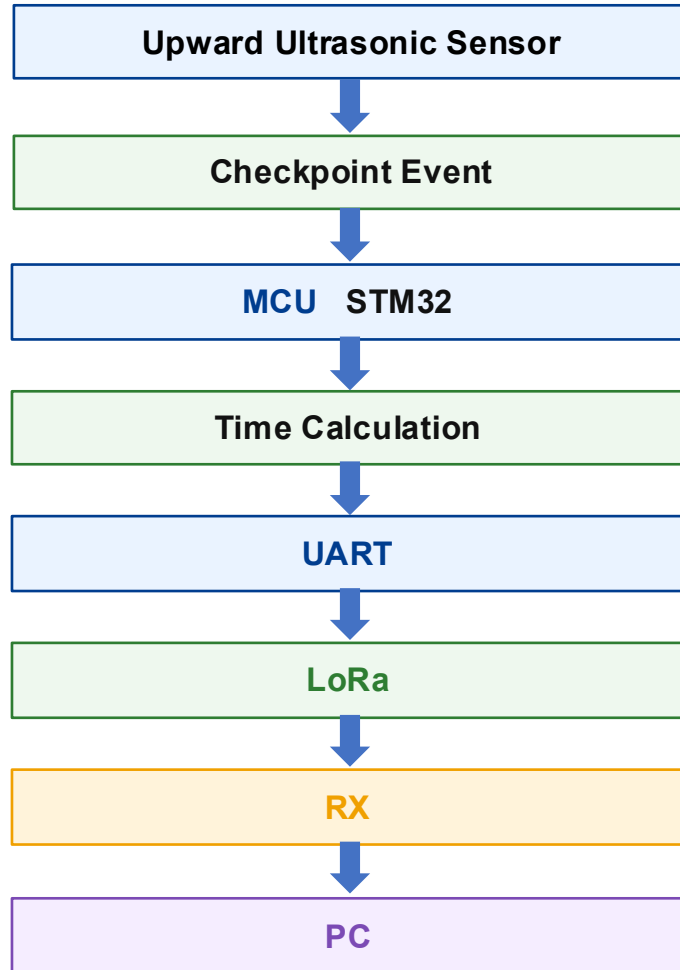


HC-SR04

Advantages

- ✓ Detects overhead structures
- ✓ Provides clear checkpoint trigger
- ✓ Simple integration with STM32

1 Final System Architecture



Event-driven checkpoint telemetry system

2 Event Handling Strategy

After checkpoint detection, the system processes the event in a fixed sequence:



Why Stop?

- ✓ Avoid repeated triggering
- ✓ Stable event processing
- ✓ Easy communication verification

Example Output

Team 15 PTSD
Duration 01:23 8:00

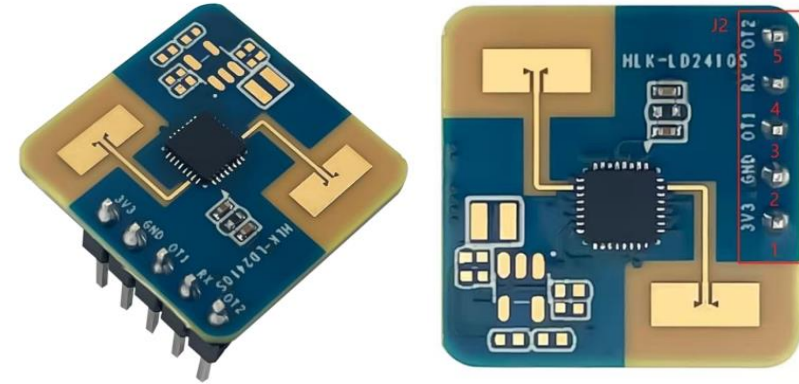
Team 15 PTSD
Duration 01:34 8:00

Radar Obstacle Avoidance Subsystem

01. Solution Selection and Core Advantages

Core Component: HLK-LD2410S 24GHz Millimeter Wave Radar

- Strong Environmental Adaptability : Unaffected by light, temperature, dust, stable outdoors.
- Abundant Resources : More comprehensive docs than CYPLUS radar.



Radar module appearance

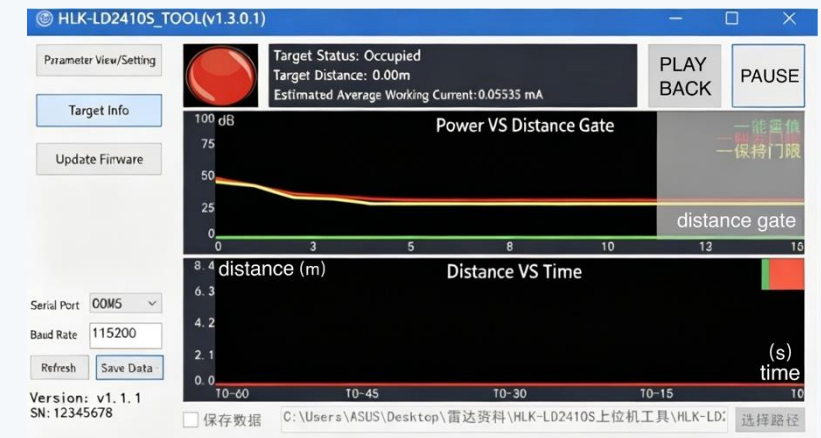
02. Standalone Radar Testing

Used manufacturer's PC software (HLK-LD2410S_TOOL) to verify radar works

The software clearly displays:

Whether a target is present, distance vs. time curve, power vs. range gate curve

Solution for Inaccuracy ($< 0.35\text{m}$) : Use "presence/absence" judgment (no precise distance needed).



PC software

03. Radar-to-MCU Communication Verification

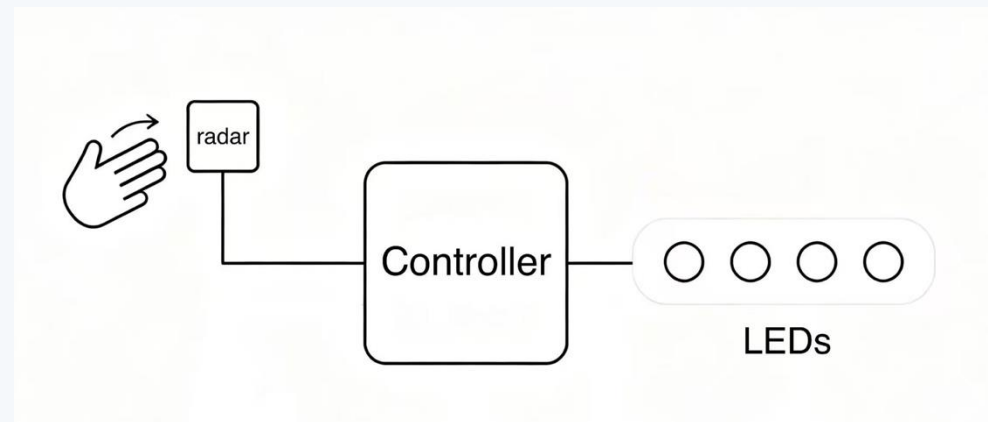
Debugging: 4 LEDs verify communication/parsing.

LED1: Radar powered on

LED2: Radar sending data to STM32

LED3: Obstacle detected

LED4: STM32 decided to turn to clear side



LEDs testing schematic diagram

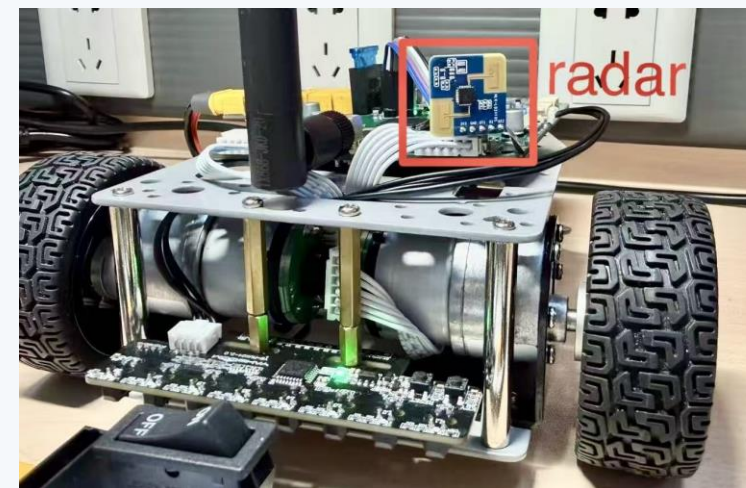
04. Achievement Summary

System Status

The radar module works stably, detecting obstacles in real-time and reliably.

Project Contribution

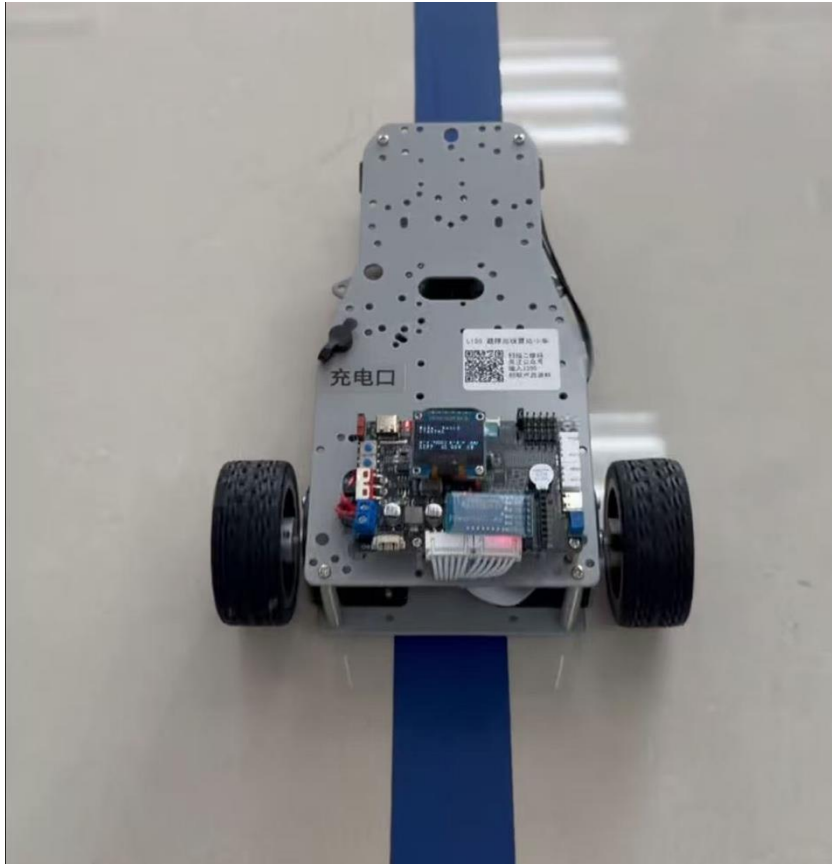
Successfully implemented autonomous obstacle avoidance, completing the project satisfactorily.



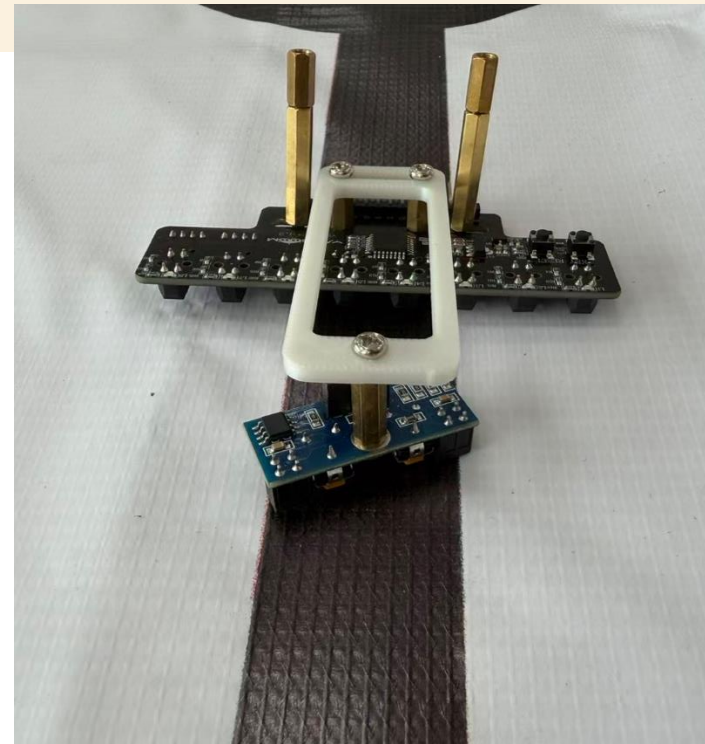
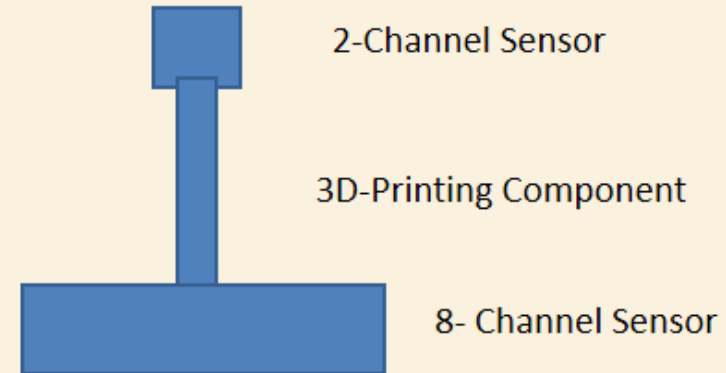
Radar installed on vehicle

Line Following Sensor

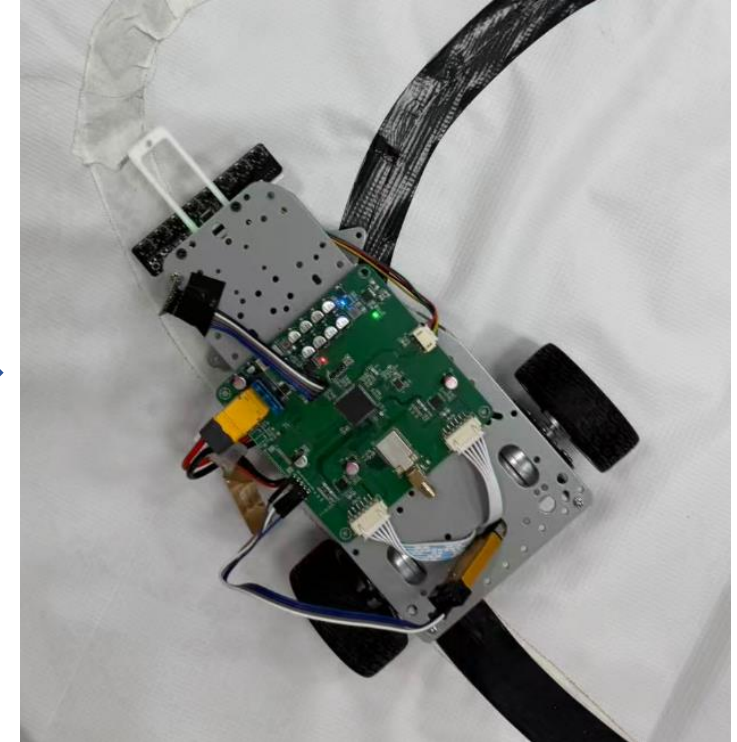
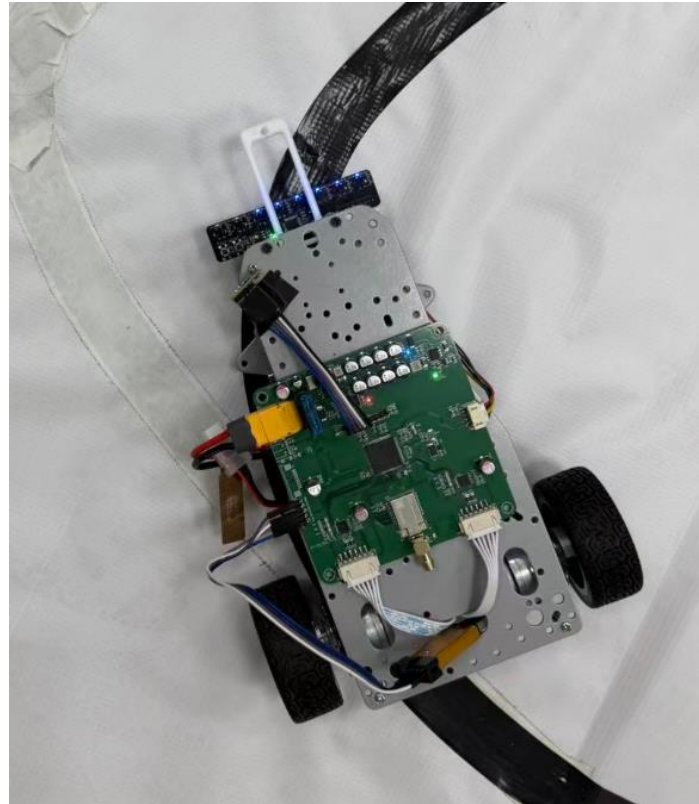
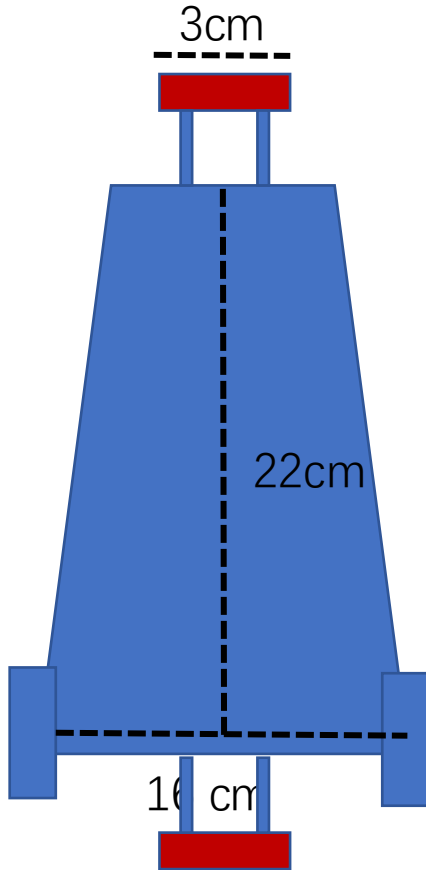
Mechanical Design & Sensor



3-wheel differential drive system

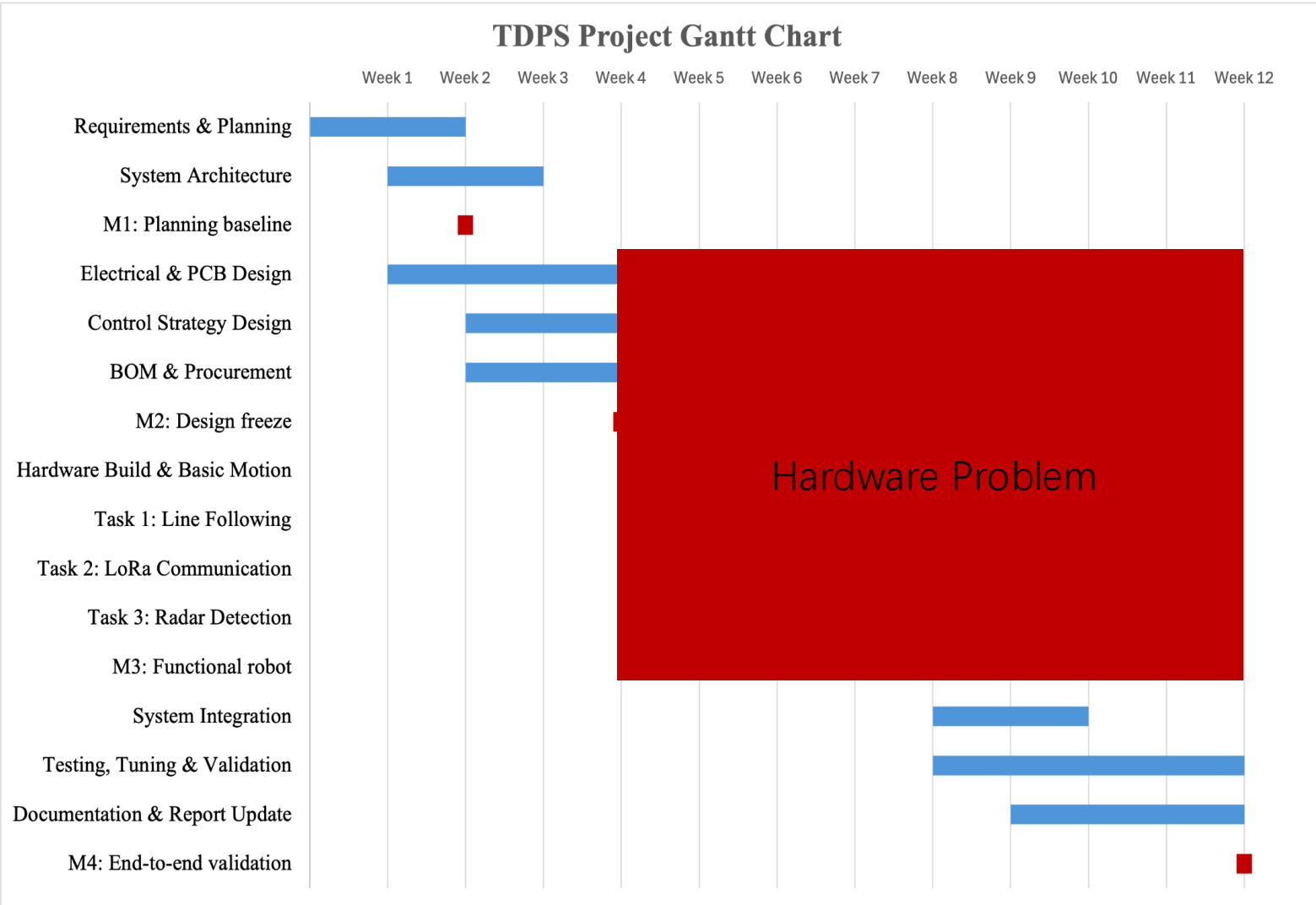


New problem



“tiny correction at the 16cm-spaced wheels caused the front sensor to swing violently across the track”

Some Problems.....



PCB Redesign.....

The interface is reversed
need to be re-welded.....

The sensor is short-circuited
and burned.....

1

System Integration

Scope / Deliverables



- PCB6 integrates power and interfaces
- Solderable platform for repeated testing

2

Demo Limitation

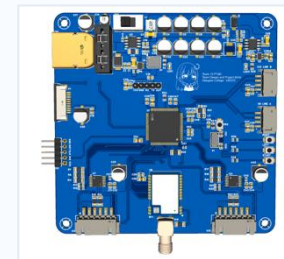
Field Validation Gap

- Final field changed sensor response
- Line reflectivity and floor material differed
- Lab-ready results still needed field validation

3

Cost-Risk Trade-off

Reliability / Debugging / TCO



- Adapter boards were fast but risky
- PCB6 improved wiring and power clarity

Key Lesson

A working prototype must be field-ready, not only lab-ready.

Future focus: adaptive threshold tuning · early on-site validation · modular PCB bring-up · fallback strategy

Robust Integration

Risk Mitigation

Power Debugging

Field Validation

Thanks for listening

Team 15

